11. Optimal Page Replacement

```
# include <stdio.h>

int main () {
	int i,j,k,n,frames,faults=0;

	printf("Enter the number of pages: ");
	scanf("%d", &n);

	int pages[n];
	printf("Enter page reference string: ");
	for (i=0;i<n;i++) {
		scanf("%d",&pages[i]);
	}

	printf("Enter the number of frames: ");
	scanf("%d", &frames);

	int frame[frames];

	for (i=0;i<frames;i++) {
		frame[i] = -1;
	}

	for (i=0;i<n;i++) {
		int found = 0;

		for (j=0;j<frames;j++) {
			if (frame[j] == pages[i]) {
				found = 1;
				break;
			}
		}

		if (!found) {
			int pos = -1, farthest = i + 1;

			for (j=0;j<frames;j++) {
				int next = -1;

				for (k=i+1;k<n;k++) {
					if (frame[j] == pages[i]) {
						next = k;
						break;
					}
				}
```

```
            if (next == -1) {
                pos = j;
                break;
            }

            if (next > farthest) {
                pos = j;
                farthest = next;
            }
        }

        if (pos == -1) {
            pos = 0;
        }

        frame[pos] = pages[i];
        faults++;

    }
}

printf("Total Page Faults: %d\n", faults);

return 0;
}
```